

SMART INDIA HACKATHON 2026



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HACKATHON
2026**

TITLE PAGE

Problem Statement ID : 26157

**Problem Statement Title : Supervisory Analytics Tool
for SOC Assessment (SAT-SA)**

Theme : Blockchain & Cybersecurity

PS Category : Software

Team ID : -

Team Name : CodeTech

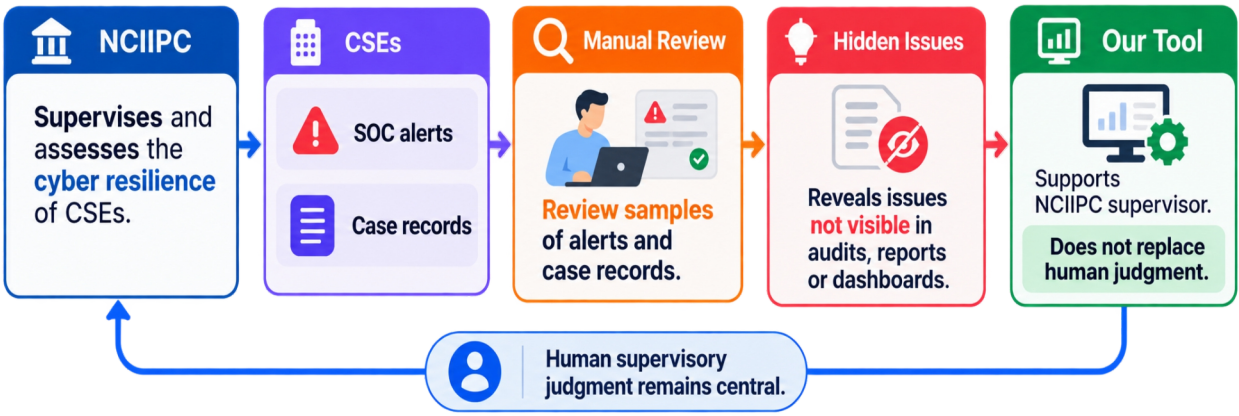


- **NCIIPC (National Critical Information Infrastructure Protection Centre)** is the national nodal agency responsible for protecting Critical Information Infrastructure (CII).
- **CSES (Critical Sector Entities)** operate critical systems and maintain their own cybersecurity operations.
- Each CSE has a **SOC (Security Operations Centre)** that handles security activities through:

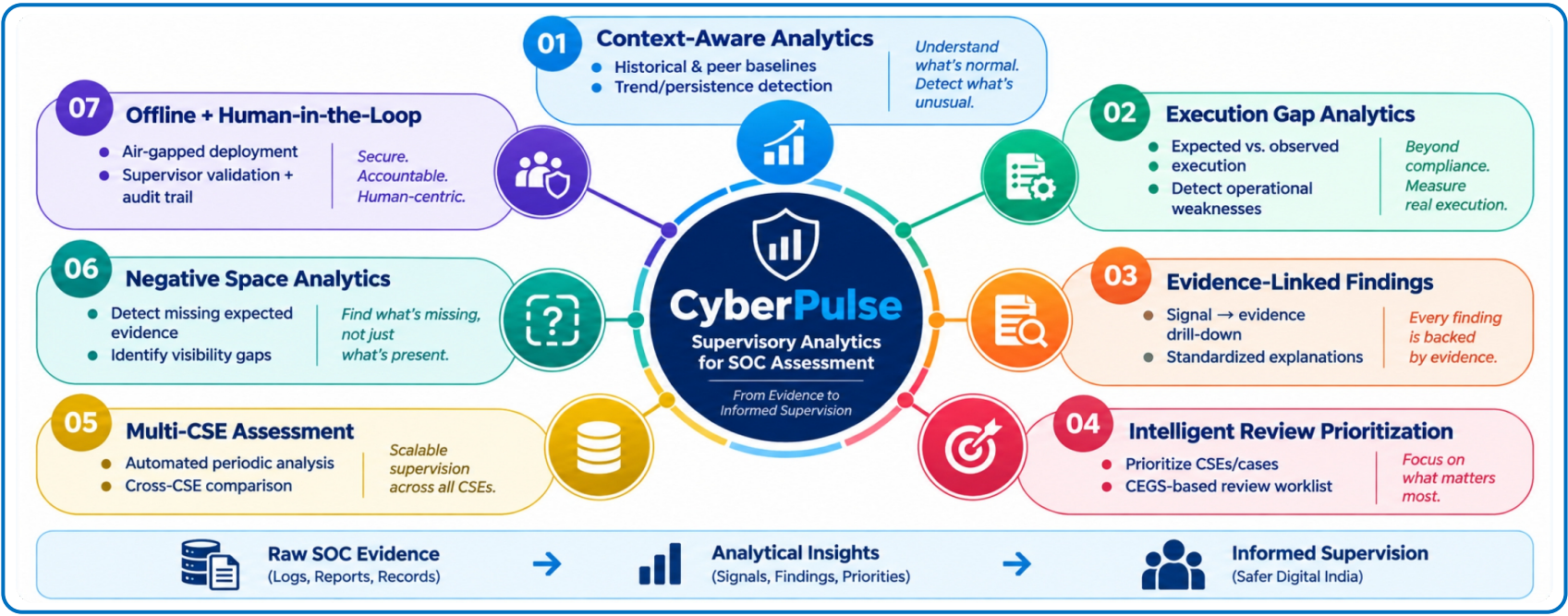


The SOC generates operational evidence such as:

- ✓ Security alert metadata
 - ✓ Case-management records
 - ✓ Investigation workflow
- ✓ Escalation records
 - ✓ Alert disposition/closure
 - ✓ Asset/system inventory



 Problems Identified	 Our Solution
 Execution Gaps - Controls/ processes appear effective, but actual execution is weak.	 Execution Gap Detection using SOC metrics and operational evidence.
 Negative Space - Expected activities/ evidence are missing.	 Negative Space Detection to identify missing investigations, escalations or monitoring evidence.
 Manual & Resource-Intensive Review - Large volumes make manual assessment difficult to scale	 Automated analytics + multi-CSE data ingestion to analyse large datasets efficiently.
 Inconsistent CSE Assessment Different formats and practices make cross-CSE comparison difficult.	 Standardized Metrics + Benchmarking Normalize evidence and compare CSE performance consistently.



TECHNOLOGY STACK

**React.js**

Dashboard & UI
Interactive dashboard &
evidence review.

**Python FastAPI**

Secure APIs & Backend
Secure APIs & Backend

**Pandas + NumPy**

Data Validation and
Processing
Cleans and prepares data.

**SciPy + scikit-learn**

Metrics & Anomaly Detection
Metrics, baselines and
anomaly detection.

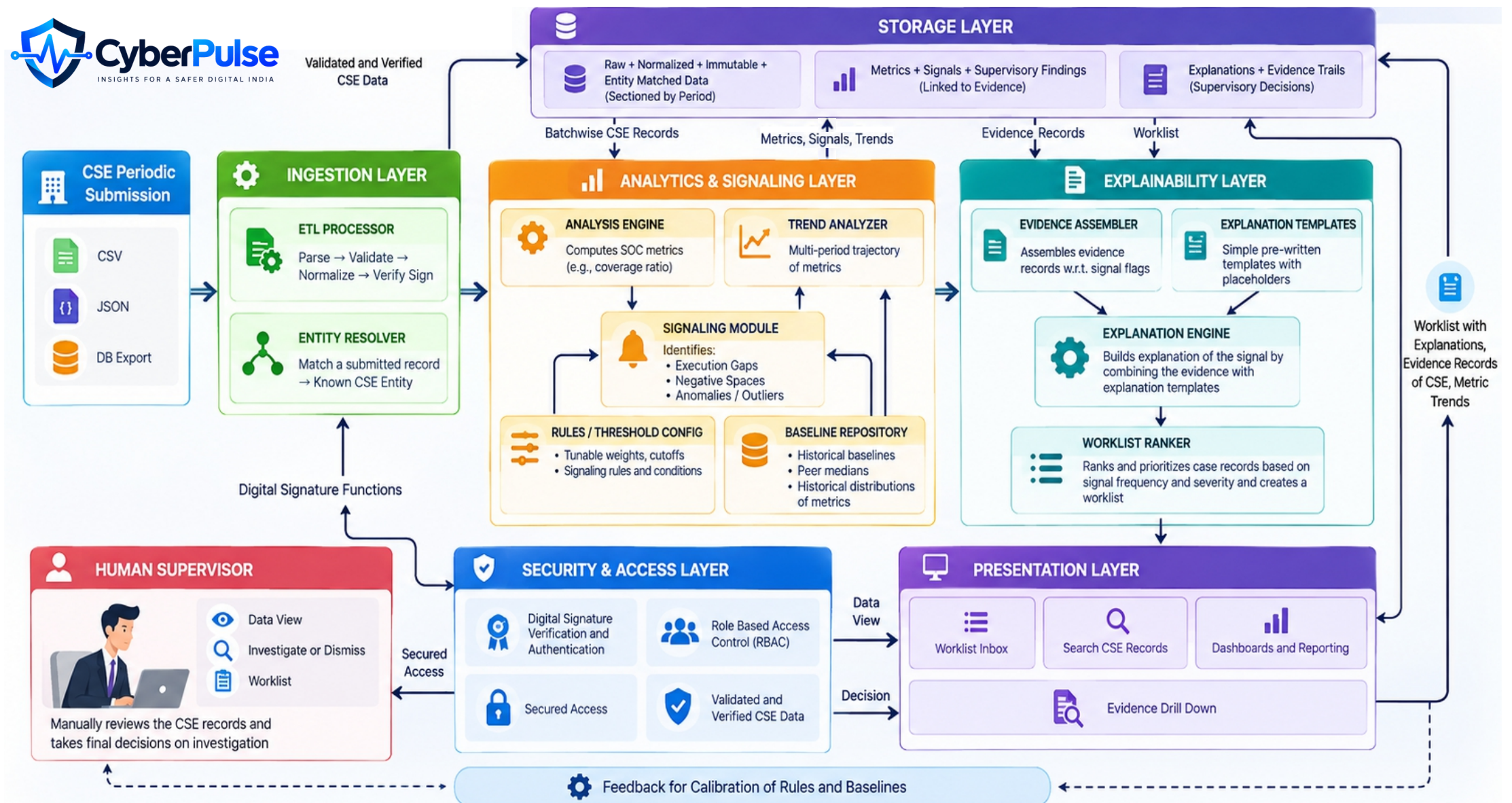
**PostgreSQL + Storage**

Data & Audit Trail
Stores data, evidence
and audit records.

**Docker + RBAC**

Secure & Reproducible
Deployment
Secure access and
consistent deployment.

SYSTEM ARCHITECTURE



**FEASIBILITY**

Can be implemented effectively

**Technically achievable****Offline and secure deployment****Fits into supervisory workflows****Cost-effective and scalable****VIABILITY**

Addresses a real and critical need

**Addresses a real and critical need****Usable by NCIIPC and CSE supervisors****Reduces manual effort and cost****Scalable across multiple sectors****SWOT ANALYSIS**

Key internal strengths and weaknesses, and external opportunities and threats

**STRENGTHS**

- Detects execution gaps and missing alerts
- Works completely offline
- Explainable and traceable results
- Scalable across multiple CSEs
- Risk-based prioritization

**WEAKNESSES**

- No real-time detection
- Depends on data quality
- May have false positives
- Data standardization changes
- Requires manual updates

**OPPORTUNITIES**

- National cybersecurity benchmarking
- Integration with global frameworks (e.g. MITRE ATT&CK)
- Internal use by CISOs
- Policy and regulatory improvements
- Local AI/NLP advancements

**THREATS**

- Metric gaming by organizations
- Manipulation of submitted data
- Evolving cyber threats
- Sensitive data leakage risk
- Resistance to adoption

**KEY CHALLENGES AND STRATEGY**

From constraints to a scalable solution

**Challenges**

Diverse data formats, data quality, and user trust.

**Strategy**

Standardized ingestion, robust validation, and explainable analytics.

**Goal**

A reliable, secure and scalable tool for stronger cybersecurity oversight.

Potential impact on the target audience
Benefits of the solution (social, economic, environmental, etc.)

#Sr. No.	Author	Title	Source	Date (Year)	Findings
1.	Vielberth, Böhm, Fichtinger, Pernul	Security Operations Center: A Systematic Study and Open Challenges	 IEEE Access	2020	Defines SOC through People, Processes, Technology and Governance; highlights challenges in SOC operations, metrics and governance.
2.	Agyepong, Cherdantseva, Reinecke, Burnap	Challenges & Performance Metrics for Security Operations Center Analysts: A Systematic Review	 Journal of Cyber Security Technology	2020	Systematic review linking SOC challenges with performance metrics; highlights limitations of volume-only metrics and need for contextual assessment.
3.	Shah, Ganesan, Jajodia, Cam	A Methodology to Measure and Monitor Level of Operational Effectiveness of a CSOC	 International Journal of Information Security	2018	Proposes operational-effectiveness measurement using baselines, thresholds and deviations to monitor deterioration and recovery.
4.	Forsberg & Frantti	Technical Performance Metrics of a Security Operations Center	 Computers & Security	2023	Provides a framework for designing defensible SOC metrics based on goals, measurability, bias, context, comparability and usability.
5.	https://ieeexplore.ieee.org/document/9296846				
6.	https://www.tandfonline.com/doi/abs/10.1080/23742917.2019.1698178				
7.	https://link.springer.com/article/10.1007/s10207-017-0365-1				
8.	https://www.sciencedirect.com/science/article/pii/S016740482300439X?via%3Dihub				